

Level 1 - Searching Foundations

1. Linear Search

```
nums = [8, 3, 7, 1, 5]

target = 7

def linear_search(nums):

    for i in range(len(nums)):

        if nums[i] == target:
            return i

    return -1

print(linear_search(nums))
```

2. Binary Search

```
nums = [1, 3, 5, 7, 9, 11, 13]
target = 11

def binary_search(nums):

    left = 0
    right = len(nums) - 1

    while left <= right:
        mid = (left + right) // 2
        if nums[mid] == target:
            return mid

        elif nums[mid] < target:
            left = mid + 1

        else:
            right = mid - 1

    return -1

print(binary_search(nums))
```